Malaysia Price Watch

Grocery price report

Moztoba Mutashim Billah November 2026

Prepared for KPDN price monitoring, a supermarket category team and an FMCG procurement team. Data from October 2025 to September 2026, analysed with PostgreSQL, SQL, Python and Power BI. Live dashboard: moz700.github.io/malaysia-price-watch/dashboard. Code and data: github.com/Moz700/malaysia-price-watch.

Summary

Households, retailers and suppliers in Malaysia set prices and plan purchases from national averages. Those averages hide where the same item costs more, and in which kind of shop. This report draws on 21 million government shelf-price records to set out a KPI scorecard for three stakeholders, the findings behind it and an action plan.

Location and shop format move prices by similar amounts. The same 265-item basket cost 9.9% (RM 390.12) more in Kuala Lumpur than in Kedah, and on identical items hypermarkets and sundry shops were 9.0% apart. For KPDN and for retailers, monitoring and pricing set by state average miss the shop-format effect, which is as large as the location effect.

Sugar and cooking oil were at or below the controlled price in 99.95% of shop-months. The gaps are in uncontrolled fresh produce and dry goods, so KPDN can move enforcement effort from controlled items to fresh produce in the dearest districts.

Wet markets charge more than hypermarkets for fresh food (index 106.5 against 97.7), and a large format beat them in 14 of 16 states. Eight everyday staples cost 12.0% more at a wet market than at a hypermarket. Supermarkets can win on price perception with fresh produce, except in Kuala Lumpur, where their fresh produce index is 115.8. The average supermarket is 12.1 points behind the cheapest chain.

Cost risk is concentrated. Three categories, holding 17% of items, caused 58% of all price increases, and fresh prices swing 8 times more than packaged goods. For procurement, fixed contracts suit packaged goods, while vegetables, onions and meat need indexed contracts and a second source.

Festive staples fell 2.4% before Chinese New Year and 5.0% before Hari Raya Aidilfitri. December was the high-risk month, with 78 price-spike weeks, 74 of them in fresh produce. In 2026, festive stock builds carried less price risk than December buying, so procurement teams and KPDN should lock fresh-staple prices before December.

1 Stakeholders and KPIs

The three stakeholders make different decisions, so each has its own KPIs (Table 1). Every KPI is calculated in SQL from the same cleaned database, and rerunning one script refreshes them all.

Table 1 Stakeholders and decisions

Who uses the report and for which decision. Their KPIs are in Table 2.

Stakeholder	Business problem	Decision supported
KPDN price monitoring	Limited inspectors, nationwide complaints	Where to monitor and enforce
Supermarket category manager	Shoppers compare formats and chains	Which categories to price against whom
FMCG procurement analyst	Input costs move without warning	Contract type and timing of purchases

Table 2 KPI scorecard

Values are for September 2026 or the full year, as stated. Alert rules are proposed thresholds for a monthly review, and the status follows the rule.

KPI	Definition	Value	Alert rule	Status
KPDN price monitoring				
Regional basket gap	Same 265-item basket, dearest vs cheapest state	9.9% (RM 390.12)	> 5%	Act
District price spread	Widest district gap inside one state, index points	11.7 (KL)	> 8	Act
Monitoring coverage	Districts with 4+ reporting shops	76.4%	< 90%	Act
Price-control adherence	Shop-months at or below the controlled price, sugar and cooking oil	99.95%	< 99%	On track
Supermarket chain, category manager				
Price position index	Average supermarket vs national median (100)	101.4	> 100	Act
Gap to price leader	Index points behind the cheapest chain (Tesco, 89.3)	12.1	> 5	Act
Items priced above market	Supermarket items above the national median	32.4%	> 25%	Act
Fresh produce index	Supermarket fresh produce vs national median	100.9; KL 115.8	> 105	Act (KL)
FMCG procurement				
Annual price change	Median item, same shops, Oct 2025 to Sep 2026	0.0%	> 3%	On track
Increase concentration	Share of all increases from the top 3 categories	58%	> 50%	Act
Fresh price volatility	Coefficient of variation of monthly prices (packaged: 0.87%)	6.92%	> 5%	Act
Spike frequency	Item-weeks more than 2 standard deviations and 2% above the prior 8 weeks	4.2%; Dec 78 weeks	Dec peak	Watch

Sources: queries q07, q09, q10, q11, q12, q13, q14, q23, q29 and q31 (Appendix A).

2 Regional prices and monitoring (KPDN)

Kuala Lumpur was the dearest state. The same 265-item basket cost RM 4,349.65 there and RM 3,959.53 in Kedah, RM 390.12 or 9.9% more (Figure 1). The five dearest are the three federal territories, Sabah and Sarawak.

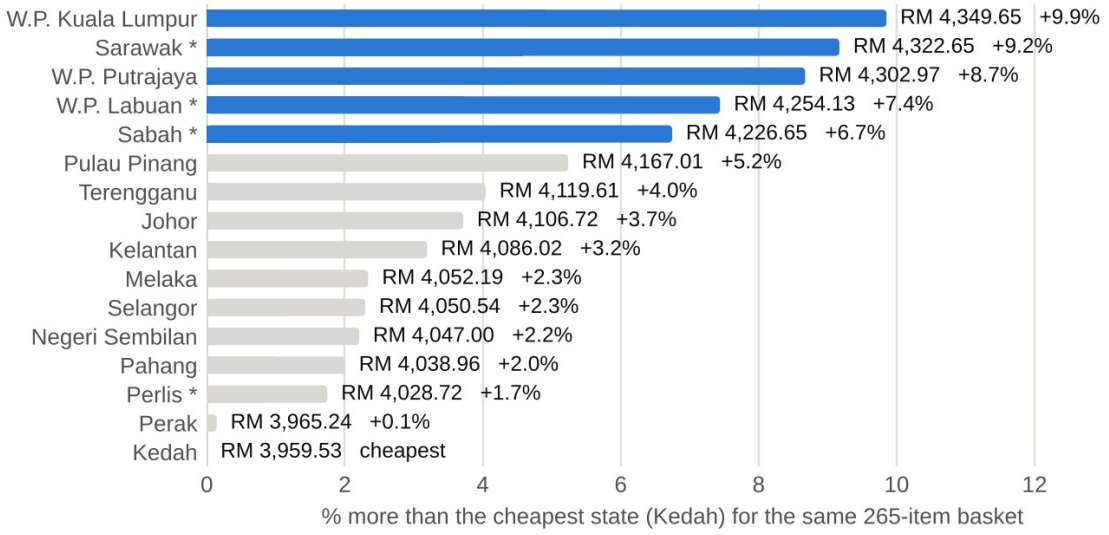
Districts in the same state differ as well. In Kuala Lumpur the district basket index runs from 100.4 in Bandar Tun Razak to 112.0 in Segambut, and five states have a district spread above 8 points (Figure 2). Nationally, the dearest districts are Limbang (113.9), Segambut (112.0) and Lawas (111.4).

Sugar and cooking oil stayed within the controlled price. Coarse sugar (ceiling RM 2.85) and packet cooking oil (RM 2.50) were at or below it in 99.95% of 34,609 shop-months (Bernama, 2024; Malay Mail, 2026). The price gaps in this report come from fresh produce and dry goods, which are not price-controlled, so inspection time is better spent there.

Coverage is thinnest in Sabah and Sarawak, two of the dearest states. In September 2026 only 76.4% of listed districts had four or more reporting shops, and 22 of the 30 districts with only one to three are in Sabah or Sarawak.

Figure 1 Cost of the 265-item basket by state, September 2026

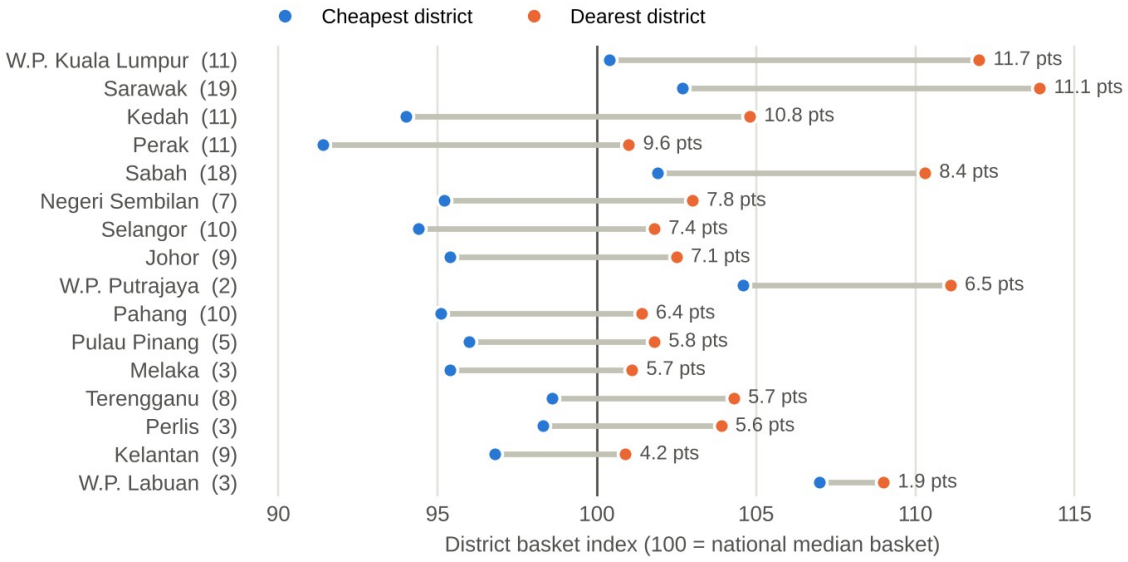
At each state's median prices. * Ten or more items not priced in the state, filled with the national median.



Source: query q09.

Figure 2 Cheapest and dearest district basket index in each state, September 2026

100 = national median basket.



Source: query q10. Gaps are calculated before rounding.

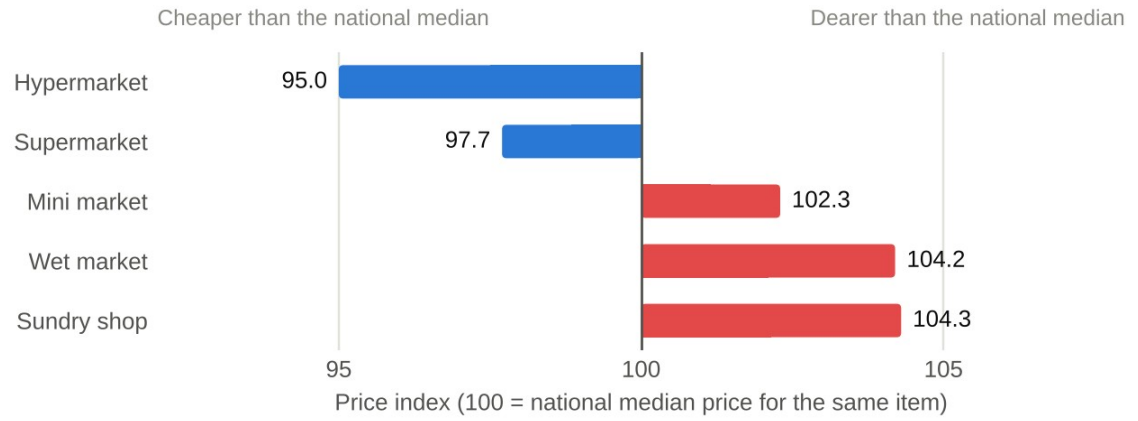
For KPDN, the findings support moving inspections from controlled items to fresh produce and dry goods in the dearest districts, and adding reporting shops in the Sabah and Sarawak districts with three or fewer. KPDN should also run a weekly vegetable price watch from November to January.

3 Price positioning by shop format (supermarket chain)

Shoppers pay most at small formats. On 181 items sold by all five formats, hypermarkets were the cheapest, at 5.0% below the national median, and sundry shops were 4.3% above it (Figure 3). One unit each of eight everyday staples cost RM 88.76 at hypermarkets and RM 99.45 at wet markets, 12.0% more (Figure 4).

Figure 3 Price index by shop type, September 2026

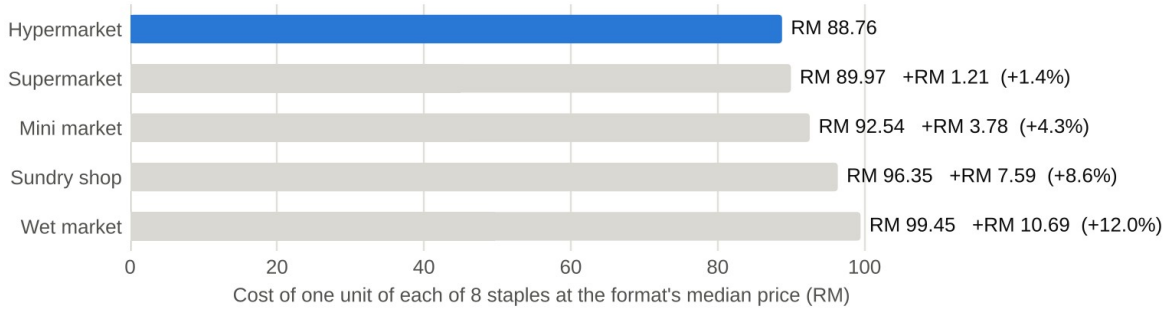
181 items sold in all five formats; 100 = national median.



Source: query q03.

Figure 4 Cost of eight everyday staples by shop type, September 2026

One unit each of chicken, eggs, onions, garlic, red chilli, rice, sugar and cooking oil, at median prices.

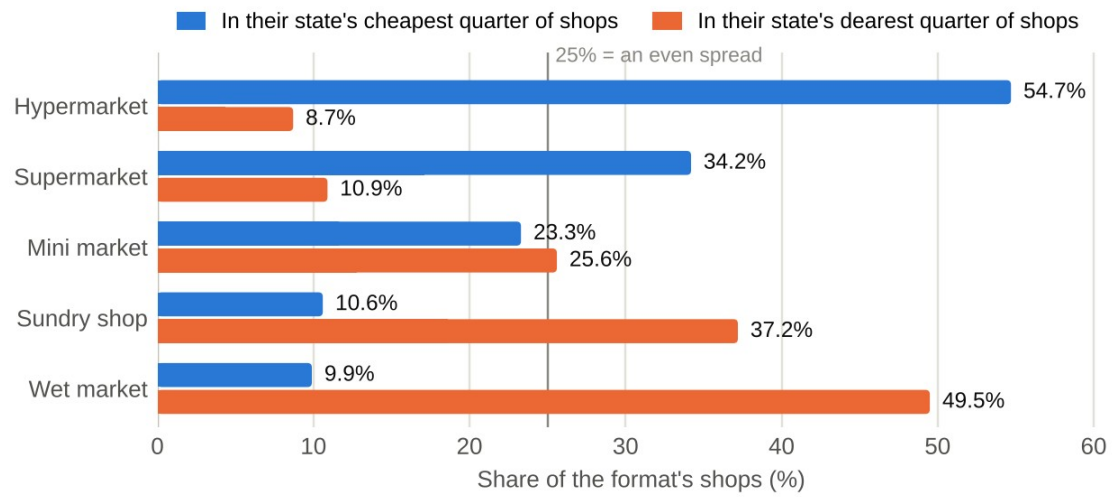


Source: query q03.

Supermarkets sit in the middle. The average supermarket scored 101.4, against 89.3 for Tesco, 89.4 for Lotus's and 90.2 for Econsave's 58 branches, which leaves it 12.1 points behind the price leaders. Within their state, 54.7% of hypermarkets fall in the cheapest quarter of shops, and half of wet markets (49.5%) fall in the dearest quarter (Figure 5).

Figure 5 Shops in the cheapest and dearest quarter of their state, by format, September 2026

Share of each format's shops.



Source: queries q11 and q19.

Branded packaged goods sit within a few points of 100 in every format, so there is little to win there. The gaps are in fresh produce and dry goods (Table 3). Wet markets are assumed to be the cheap option for fresh food, yet they averaged 106.5 against 97.7 at hypermarkets and lost to a large format in 14 of 16 states. In Kuala Lumpur, though, supermarket fresh produce scored 115.8, against 100.8 at hypermarkets.

Table 3 Price index by product group and format, September 2026

Shops pricing 30 or more items. Below 100 is cheaper than the national median.

Product group	Items	Hypermarket	Supermarket	Mini market	Wet market	Sundry shop
Fresh produce	115	97.7	100.9	109.4	106.5	113.0
Dry goods	22	94.4	99.6	107.1	119.4	112.8
Packaged goods	109	98.2	100.6	100.2	104.0	102.7
Milk and baby products	25	99.2	101.8	100.7	104.3	105.3
Household and hygiene	27	97.5	100.0	100.3	102.9	106.1
Beverages	31	98.6	101.3	99.9	101.1	103.1

Source: query q12.

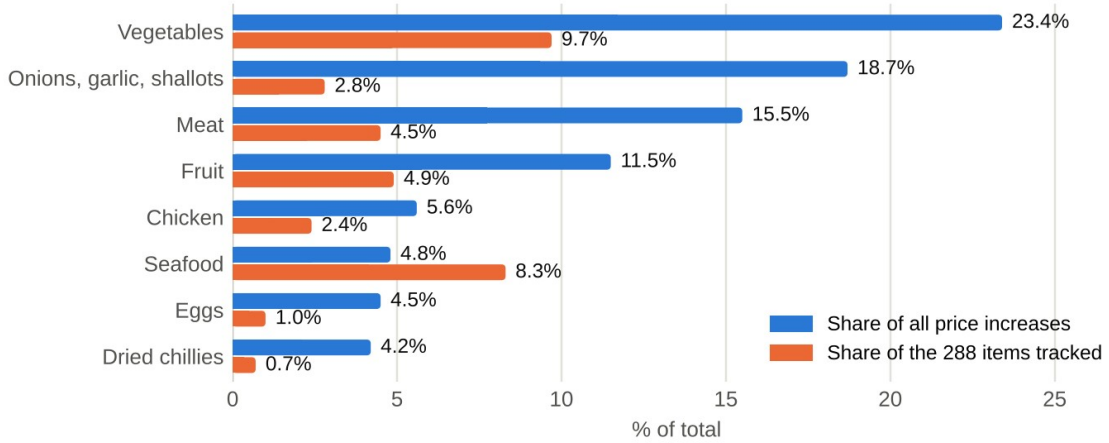
The category manager should benchmark fresh produce and dry goods against Tesco, Lotus's and Econsave rather than independent shops, starting with a review of Kuala Lumpur fresh produce pricing. The gap to the price leader is worth tracking monthly.

4 Cost risk (FMCG procurement)

Most prices were flat. In the same shops, 84% of 288 items moved less than 5% over the year, and the median item did not change. Vegetables, onions and garlic, and meat produced 58% of all increases with 17% of the items (Figure 6). Tomatoes rose 69.4%.

Figure 6 Share of price increases and of items, by category

288 items in the same shops, October 2025 to September 2026.

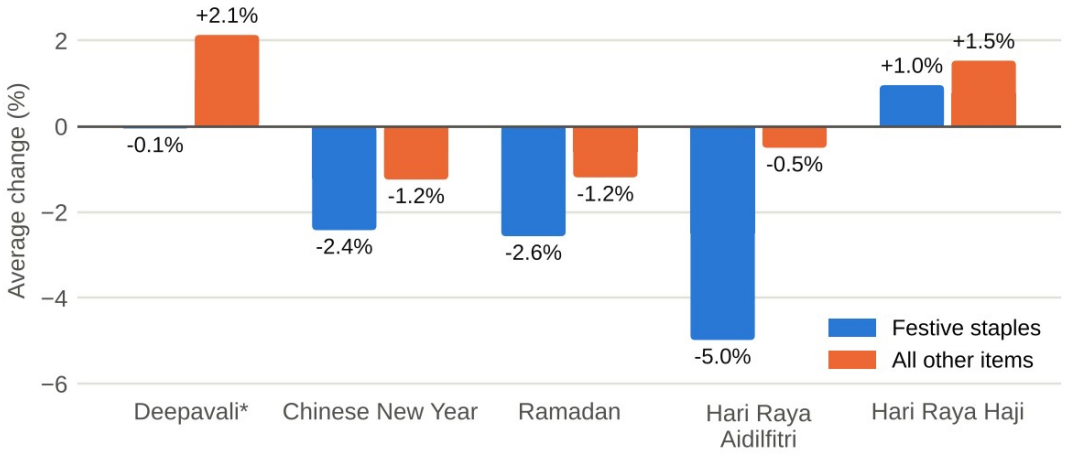


Source: query q14.

Prices fell before festivals and spiked in December. In the same shops, festive staples fell 2.4% before Chinese New Year and 5.0% before Hari Raya Aidilfitri (Figure 7). Spikes were less frequent before festivals (2.83% of item-weeks) than in other weeks (4.52%). December 2025 had 78 spike weeks, 74 of them in fresh produce (Figure 8).

Figure 7 Average price change before each festival, same shops

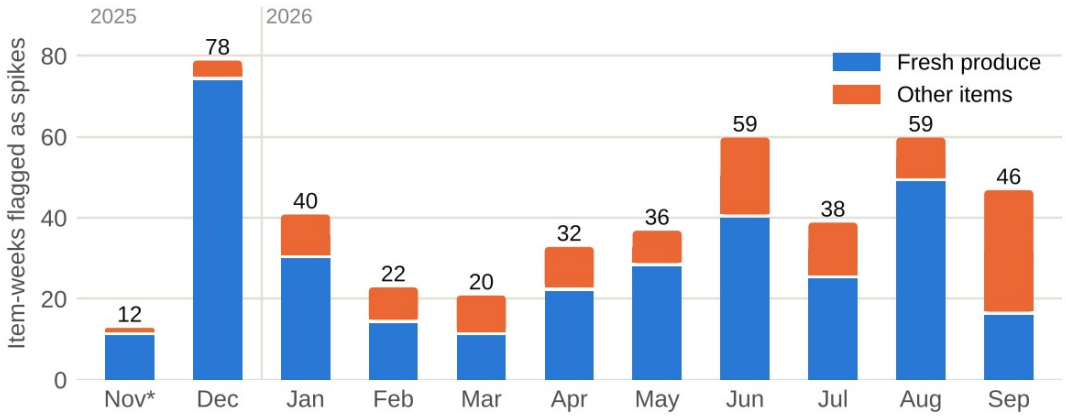
The 14 days before each festival against days 42 to 15 before it. * Deepavali baseline covers only 1 to 5 October.



Source: query q21.

Figure 8 Item-weeks flagged as price spikes, by month

* November 2025 is partial.



Source: query q23.

The budget risk is month-to-month volatility. Measured as the coefficient of variation of monthly prices, fresh produce varied by 6.92% on average, against 0.87% for packaged goods. Red chilli (cili akar) ranged from RM 13.99 to RM 32.00 a kilogram across the year's monthly medians, 2.3 times from low to high (Table 4). Sugar and cooking oil, both price-controlled, never moved, while chicken, eggs and chilli did (Figure 9).

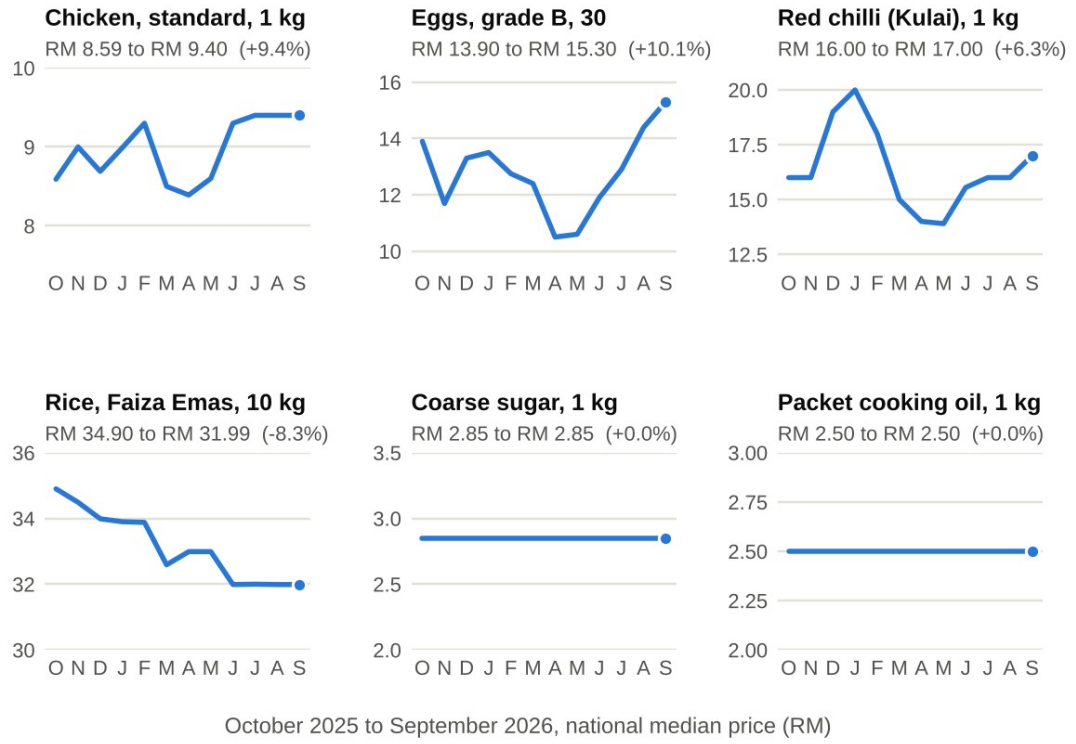
Table 4 The most volatile items

Lowest and highest monthly national median and the coefficient of variation, October 2025 to September 2026.

Item (1 kg)	Lowest (RM)	Highest (RM)	Variation
Red chilli (cili akar)	13.99	32.00	33.4%
Shallots, rose variety, imported (India)	5.70	12.99	32.3%
Red grapes, seeded	10.50	22.90	28.3%
Tomato	4.00	9.49	26.8%
Seedless red watermelon	1.99	4.49	22.1%

Source: query q29.

Figure 9 National median price of selected staples by month, October 2025 to September 2026



Source: queries q17 and q20.

Procurement should put packaged goods on fixed-price contracts and use indexed contracts with a second source for vegetables, onions and meat. Fresh-staple prices should be locked in October and November, ahead of the December spike window.

5 Recommended actions

Table 5 Actions, owners and targets

Each action, its owner, the KPI that tracks it and a proposed target.

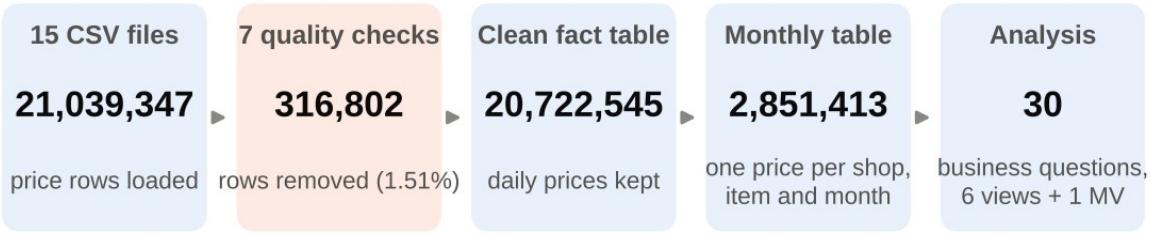
#	Action	Owner	KPI tracked	Proposed target	When
1	Move inspections from controlled items to fresh produce and dry goods in the ten dearest districts	KPDN enforcement	District price spread	Below 8 points	Next quarter
2	Add reporting shops in the 22 Sabah and Sarawak districts with one to three	KPDN PriceCatcher team	Monitoring coverage	90% of districts	Within 6 months
3	Run a weekly vegetable price watch from November to January	KPDN, procurement	Spike frequency	Alerts within one week	Every November
4	Benchmark fresh produce and dry goods against Tesco, Lotus's and Econsave	Category manager	Gap to price leader	Below 5 points	Monthly
5	Review Kuala Lumpur supermarket fresh produce pricing	Category manager, KL	Fresh produce index (KL)	Below 105	Within 3 months
6	Fixed contracts for packaged goods; indexed contracts and a second source for vegetables, onions and meat	Procurement	Fresh price volatility	Budget variance within 5%	Next contract cycle
7	Lock fresh-staple prices with suppliers in October and November	Procurement	Spike frequency	No December buying at spike prices	Yearly

The KPIs are built to refresh monthly. Loading the new PriceCatcher files and rerunning the build script updates the reporting views, and these feed a seven-page Power BI report: an overview, one page for each stakeholder, a shop ranking, an item lookup and a page on the data.

6 Data and method

Three method choices sit behind the numbers. A wet market reports each item about 15 times a month and a sundry shop about 3, which is why every comparison uses one price per shop. Because shops join and leave during the year, changes over time compare the same shops; a naive comparison would show imported Pakistani onions up 3.7% instead of 35.2%. Most small shops are surveyed on Mondays, so spikes are measured weekly.

Figure 10 Build pipeline
Row counts at each stage of the build.



Source: build run log.

Table 6 Build stages
What was done at each stage and the evidence of quality.

Stage	What was done	Result
Data	12 monthly PriceCatcher files, item and shop lookups, DOSM CPI by state	21,039,347 raw price records; 2,262 shops and 338 items after cleaning; 16 states
Data quality	7 documented checks: impossible prices, duplicates, unknown items and shops, outliers, shop churn	1.51% removed; 99% of that from 17 item codes missing in the published list
Data model	PostgreSQL star schema with a daily fact table and a monthly one-price-per-shop table	Each shop counts once in every comparison
Analysis	30 stakeholder questions and a KPI query (q31) in SQL: CTEs, window functions, GROUPING SETS, matched-shop comparisons	The KPIs and findings in this report
Validation	PriceCatcher food prices compared with DOSM's food CPI by state	Correlation 0.60 over 160 state-months
Performance	Composite index; materialized view for the weekly spike table	2.97 s to 0.09 s; 52.5 s to 0.05 s
Delivery	6 reporting views plus 1 materialized view for Power BI; full rebuild script	Refreshable monthly in about 15 minutes

7 Limitations

PriceCatcher is collected for price surveillance, and inflation is measured by DOSM's CPI (Ministry of Domestic Trade and Cost of Living & Department of Statistics Malaysia, 2026), which this report uses only as a cross-check. Every item has equal weight, so the baskets compare like-for-like prices rather than household spending. Format and location comparisons use September 2026, although the queries run for any month, and the data does not record freshness or bargaining.

References

Bernama. (2024, August 14). KPDN seizes 213kg of overpriced sugar in Perak supermarket. https://www.bernama.com/en/news.php?id=2328972

Department of Statistics Malaysia. (2026). *Monthly CPI by state & division (2-digit)* [Data set]. OpenDOSM. https://open.dosm.gov.my/data-catalogue/cpi_state

Malay Mail. (2026, July 16). PAC: RM10.879b cooking oil subsidy failed to fully reach intended recipients. https://www.malaymail.com/amp/news/malaysia/2026/07/16/pac-rm10879b-cooking-oil-subsidy-failed-to-fully-reach-intended-recipients/227800

Ministry of Domestic Trade and Cost of Living, & Department of Statistics Malaysia. (2026). *PriceCatcher: Transactional records* [Data set]. data.gov.my. https://data.gov.my/data-catalogue/pricecatcher

Appendix A The SQL analyses behind the KPIs

Each analysis is one SQL file in the project repository, written for one stakeholder (A KPDN, B category manager, C procurement). q31 adds the price-control adherence and monitoring coverage KPIs.

Table 7 Analysis index

What each query answers and the SQL techniques it uses.

#	For	Business question	SQL techniques
q01	A	National median price of each item, September 2026	JOIN, WHERE, GROUP BY, PERCENTILE_CONT … WITHIN GROUP
q02	A	Most and least expensive state for the 10 most common items	CTEs, JOIN, AVG, ORDER BY, LIMIT, RANK()
q03	B	How prices differ by premise type for the same item	GROUP BY two columns, HAVING, FILTER inside an aggregate
q04	B	Items with the biggest gap between the cheapest and dearest shop type	CTE, MAX − MIN, ARRAY_AGG
q05	C	How many items and premises does each state have? Is coverage balanced?	COUNT(DISTINCT), share of total with SUM() OVER ()
q06	A	Districts with the fewest reporting premises	GROUP BY, HAVING (filters groups, after counting)
q07	B	How many items does each shop type price above the national median?	CASE inside SUM, the FILTER clause
q08	C	Which item categories have the widest price range across Malaysia?	STDDEV, AVG, PERCENTILE_CONT at 0.1 and 0.9
q09	A	What does an "essentials basket" cost in each state?	multi-step CTE, CROSS JOIN, LEFT JOIN, COALESCE
q10	A	Which district has the cheapest basket within each state?	CTE + correlated subquery in WHERE, TEMP TABLE
q11	B	Premise price index: which shops are cheapest overall?	CTE, ratio metric, HAVING
q12	B	Premise price index by premise type and state	CTE + GROUP BY, FILTER to pivot types into columns
q13	C	Which items changed price most between October 2025 and September 2026?	CTE with two periods, JOIN on two keys, % change
q14	C	Are price increases concentrated in a few categories?	CTE, share of total with SUM() OVER (), running share
q15	B	Rank premises by price for each item within each district	RANK() OVER (PARTITION BY …)
q16	B	For each item and district, which premise is cheapest most often?	DENSE_RANK + COUNT, ROW_NUMBER to keep the top row per group

#	For	Business question	SQL techniques
q17	C	Month-on-month median price change for each item	LAG() OVER (PARTITION BY … ORDER BY …)
q18	C	Rolling 4-week average price for five key staples	AVG() OVER (ROWS BETWEEN 3 PRECEDING AND CURRENT ROW)
q19	A	Split premises into price quartiles within each state	NTILE(4)
q20	C	Running cumulative price change since October 2025, per item	SUM() OVER (PARTITION BY … ORDER BY …) = running total
q21	C	Do festive staples get dearer in the 2 weeks before each festival?	CTE + date windows built from a festival table, CROSS JOIN
q22	A	Which states see the biggest festive price spikes?	window function (RANK) over a GROUP BY result
q23	C	Detect price spikes: weeks far above an item's recent normal	AVG() and STDDEV() OVER (ROWS BETWEEN 8 PRECEDING AND 1 PRECEDING)
q24	B	How often does each premise change its price on the same item?	LAG + COUNT of changes, then averages by group
q25	B	First and latest price per premise and item over the year	FIRST_VALUE / LAST_VALUE, window frames, ROW_NUMBER to keep one row per group
q26	A	One summary table: by state, by premise type, and grand total	GROUPING SETS, GROUPING()
q27	All	Sense check: do PriceCatcher food prices move with CPI food (division 01)?	JOIN to a reference table, LN/EXP, CORR()
q28	B	Premises that stopped reporting for a month or more	generate_series + anti-join, LATERAL
q29	C	Which items have the most stable prices?	STDDEV ÷ AVG, HAVING on a count of months
q30	All	The three most important findings, as one small report table	chained CTEs, UNION ALL, string building with || and TO_CHAR
q31	A	KPI checks: price-control adherence and monitoring coverage	CASE inside AVG, UNION ALL, CTEs, COUNT(DISTINCT)